1 Classical mechanics

1.1 Lagrangian formulation

$\boxed{1.}$ The Lagrangian of the system consists of three terms: the kinetic energy T_{kin}, the potential energy of the vertical springs $V_{\parallel}$ and the potential energy of the horizontal springs $V_{\perp}$:

$$L = T_{\text{kin}} - V_{\parallel} - V_{\perp}$$

The kinetic term is given by

$$T_{\text{kin}} = \sum_{j,k=1}^{N} \frac{1}{2}\mu\dot{q}_{jk}^2,$$

the vertical springs' potential energy can be written as

$$V_{\parallel} = \sum_{j,k=1}^{N} \frac{1}{2}\lambda q_{jk}^2.$$

The potential energy of the horizontal springs can be calculated by first summing over all springs in the x direction and then in the y direction:

$$V_{\perp} = \sum_{k=1}^{N}\sum_{j=1}^{N} \frac{1}{2}\kappa(q_{j+1\,k} - q_{jk})^2 + \sum_{j=1}^{N}\sum_{k=1}^{N} \frac{1}{2}\kappa(q_{j\,k+1} - q_{jk})^2$$

Overall, we obtain the following formula for the Lagrangian:

$$L = \sum_{j,k=1}^{N} \frac{1}{2}\left[\mu\dot{q}_{jk}^2 - \lambda q_{jk}^2 - \kappa(q_{j+1\,k} - q_{jk})^2 - \kappa(q_{j\,k+1} - q_{jk})^2\right]$$

$\boxed{2.}$ From the Lagrange equations of the second kind, we derive the following equations of motion for each q_{jk}:

$$\mu\ddot{q}_{jk} + \lambda q_{jk} + \kappa\left[4q_{jk} - q_{j+1\,k} - q_{j-1\,k} - q_{j\,k+1} - q_{j\,k-1}\right] = 0$$

$\boxed{3.}$ Using the symmetry of the model, specifically translational invariance with periodic boundary conditions, we search for the solution to the equations of motion in the form of a discrete Fourier transform

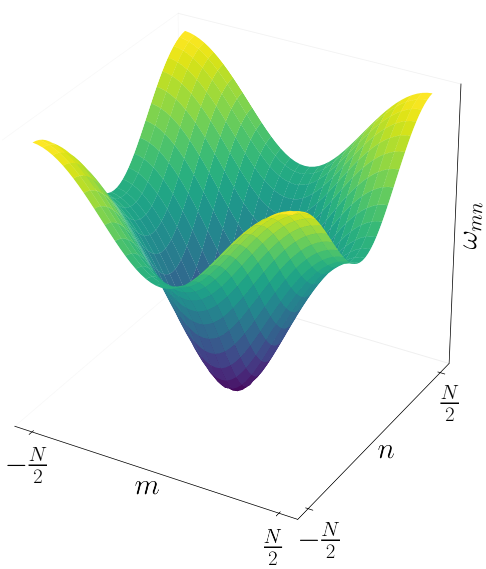
$$q_{jk}(t) = \sum_{m,n=1}^{N} \left(c_{mn} \exp\left[\frac{2\pi i}{N}(mj+nk) - i\omega_{mn}t\right] + c^*_{mn} \exp\left[-\frac{2\pi i}{N}(mj+nk) + i\omega_{mn}t\right] \right),$$

where the complex conjugate part of the solution ensures the reality of q_{jk}.

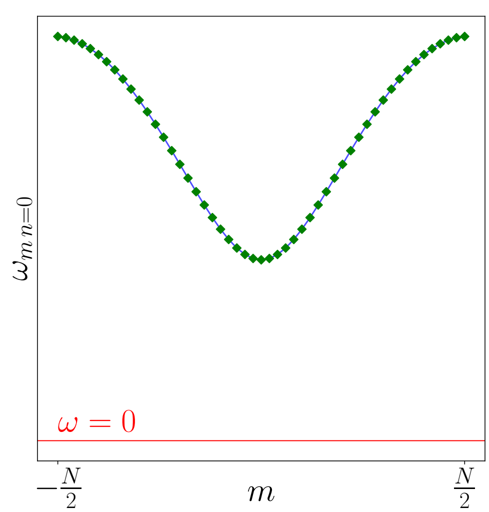
$\boxed{4.}$ By plugging this solution into the equations of motion, we obtain the following equation

$$\sum_{m,n=1}^{N} c_{mn} \exp\left[\frac{2\pi i}{N}(mj+nk) - i\omega_{mn}t\right] \left[-\mu\omega_{mn}^2 + \lambda + \kappa \left[4 - \exp\left(\tfrac{2\pi i m}{N}\right) - \exp\left(-\tfrac{2\pi i m}{N}\right) - \exp\left(\tfrac{2\pi i n}{N}\right) - \exp\left(-\tfrac{2\pi i n}{N}\right)\right]\right] + \text{c.c.} = 0.$$

For the left-hand side to be equal to zero for all j, k, and t, the last term (the last bracket) of the left-hand side must equal zero, giving us, after a few algebraic manipu-



(a)



(b)

Figure 1: This graph shows the dispersion relation ω_{mn}. The first graph (left) shows the full dependence on both m and n, while the second graph (right) shows a slice for $n = 0$. The graphs were plotted for $N = 50$, $\mu = 1$, $\lambda = 1$, and $\kappa = 1$.

lations, the dispersion relation:

$$\omega_{mn}^2 = \frac{\lambda}{\mu} + \frac{4\kappa}{\mu}\sin^2\left(\frac{\pi m}{N}\right) + \frac{4\kappa}{\mu}\sin^2\left(\frac{\pi n}{N}\right)$$

Given that the $\sin^2$ functions are always non-negative, and assuming that $\lambda, \kappa, \mu \geq 0$, the only case in which ω_{mn}, and therefore the energy, can be zero is when $\lambda = 0$. In addition, $q_{jk}(t) \equiv 0\ \forall j, k$ trivially solves the classical equations of motion with zero energy.

1.2 Hamiltonian formulation

$\boxed{1.}$ Firstly, we define the canonical momenta as

$$p_{jk} \coloneqq \frac{\partial L}{\partial \dot{q}_{jk}} \equiv \mu \dot{q}_{jk}.$$

The Hamiltonian is then given by the Legendre transform of the Lagrangian. We obtain the expression:

$$\begin{aligned} H &= \sum_{m,n=1}^{N} \frac{\partial L}{\partial \dot{q}_{mn}}\dot{q}_{mn} - L \quad \Bigg|_{\dot{q}_{jk}=\frac{1}{\mu}p_{jk}} \\ &= \sum_{j,k=1}^{N} \frac{1}{2}\left[\frac{1}{\mu}p_{jk}^2 + \lambda q_{jk}^2 + \kappa(q_{j+1\,k} - q_{jk})^2 + \kappa(q_{j\,k+1} - q_{jk})^2\right] \end{aligned}$$

Finally, we define the canonical Poisson brackets as

$$\{f, g\} \coloneqq \sum_{j,k=1}^{N}\left[\frac{\partial f}{\partial q_{jk}}\frac{\partial g}{\partial p_{jk}} - \frac{\partial f}{\partial p_{jk}}\frac{\partial g}{\partial q_{jk}}\right]$$

It is then straightforward to prove that they satisfy the following relations:

$$\{q_{jk}, q_{mn}\} = \{p_{jk}, p_{mn}\} = 0, \quad \{q_{jk}, p_{mn}\} = \delta_{jm}\delta_{kn}$$

$\boxed{2.}$ Similarly to the previous exercise, the discrete Fourier transform, respecting the periodic boundary condition, will be the key to finding the desired set of modes

(a, a^*) which diagonalizes the Hamiltonian. Firstly, we define the canonical conjugate momenta and coordinates in the Fourier space:

$$\tilde{q}_{jk} = \sum_{m,n=1}^{N} q_{mn} \exp\left[-\frac{2\pi i}{N}(jm + kn)\right]$$
$$\tilde{p}_{jk} = \sum_{m,n=1}^{N} p_{mn} \exp\left[-\frac{2\pi i}{N}(jm + kn)\right]$$

Using the dispersion relation from the first exercise, we can rewrite the Hamiltonian in the following form

$$H = \frac{1}{N^2} \sum_{j,k=1}^{N} \left[\frac{1}{2\mu}\tilde{p}_{jk}^2 + \frac{1}{2}\mu\omega_{jk}^2\tilde{q}_{jk}^2\right].$$

Finally, we define the quasiparticle modes a and a^* as

$$a_{jk} = \frac{1}{\sqrt{2\mu\omega_{jk}}}\left[\mu\omega_{jk}\tilde{q}_{jk} + i\tilde{p}_{jk}\right] = \frac{1}{\sqrt{2\mu\omega_{jk}}} \sum_{m,n=1}^{N} \exp\left[-\frac{2\pi i}{N}(jm + kn)\right]\left[\mu\omega_{jk}q_{mn} + ip_{mn}\right]$$
$$a_{jk}^* = \frac{1}{\sqrt{2\mu\omega_{jk}}}\left[\mu\omega_{jk}\tilde{q}_{jk}^* - i\tilde{p}_{jk}^*\right] = \frac{1}{\sqrt{2\mu\omega_{jk}}} \sum_{m,n=1}^{N} \exp\left[+\frac{2\pi i}{N}(jm + kn)\right]\left[\mu\omega_{jk}q_{mn} - ip_{mn}\right],$$

which allows us to write the Hamiltonian in the following form:

$$H = \frac{1}{N^2} \sum_{j,k=1}^{N} \omega_{jk} a_{jk}^* a_{jk}$$

$\boxed{3.}$ After some mostly algebraic work (which I omit for clarity), we can derive the following Poisson brackets for the quasiparticle modes:

$$\{a_{jk}, a_{mn}\} = \{a_{jk}^*, a_{mn}^*\} = 0, \qquad \{a_{jk}, a_{mn}^*\} = -iN^2\delta_{jm}\delta_{kn}$$

Using these Poisson brackets, we can derive the following Poisson bracket:

$$\{a_{jk}, H\} = \frac{1}{N^2}\sum_{m,n=1}^{N} \omega_{jk}\{a_{jk}, a_{mn}^* a_{mn}\} = \frac{1}{N^2}\sum_{m,n=1}^{N} \omega_{jk}\left[\{a_{jk}, a_{mn}^*\}a_{mn} + a_{mn}^*\{a_{jk}, a_{mn}\}\right]$$
$$= \frac{1}{N^2}\sum_{m,n=1}^{N} \omega_{jk}\left[-iN^2\delta_{jm}\delta_{kn}a_{mn} + a_{mn}^* \cdot 0\right] = -i\omega_{jk}a_{jk}$$

By taking the complex conjugate of the previous equation, we automatically obtain $\{a^*_{jk}, H\} = i\omega_{jk}a^*_{jk}$.

Using the canonical equations of motion, we obtain

$$\dot{a}_{jk} = \{a_{jk}, H\} = -i\omega_{jk}a_{jk}, \qquad \dot{a}^*_{jk} = \{a^*_{jk}, H\} = i\omega_{jk}a^*_{jk}.$$

These equations can be easily solved:

$$a_{jk}(t) = \alpha_{jk}\exp\left[-i\omega_{jk}t\right], \qquad a^*_{jk}(t) = \alpha^*_{jk}\exp\left[i\omega_{jk}t\right]$$

When we compare the solution obtained using the Hamiltonian formalism with the one derived from the Lagrangian formalism, by applying the inverse Fourier transform and solving for q_{jk}, we see that the coefficients α_{jk} and c_{jk} differ only in normalization:

$$\alpha_{jk} = N^2\sqrt{2\mu\omega_{jk}}c_{jk}$$

2 Continuum limit of the lattice model

$\boxed{1.}$ Firstly, we derive the formulae that allow us to transition from differences and sums to derivatives and integrals. For the derivatives, it is straightforward to see that

$$q_{j+1\,k} - q_{j\,k} \equiv \varphi_{[(j+1)r,kr]} - \varphi_{[jr,kr]} = \frac{\varphi_{[(j+1)r,kr]} - \varphi_{[jr,kr]}}{r}\,r \approx r\,\partial_x\varphi([jr,kr]).$$

For the sums, we introduce the area element r^2, which allows us to transition to integrals in the continuous limit:

$$\sum_{j,k=1}^{N} f_{jk}r^2 \to \int \mathrm{d}\boldsymbol{x}\, f(\boldsymbol{x}) \quad \Longrightarrow \quad \sum_{j,k=1}^{N} f_{jk} \to \frac{1}{r^2}\int \mathrm{d}\boldsymbol{x}\, f(\boldsymbol{x})$$

Overall, using the obtained formulae, we can rewrite the Lagrangian in terms of the smooth field φ:

$$L = \int \mathrm{d}\boldsymbol{x}\,\frac{1}{2}\left[\frac{\mu}{r^2}\dot{\varphi}^2 - \frac{\lambda}{r^2}\varphi^2 - \kappa\left(\boldsymbol{\partial}\varphi\right)^2\right],$$

where we have simplified the notation $\varphi \equiv \varphi(\boldsymbol{x})$ and defined $\boldsymbol{\partial} \coloneqq [\partial_x, \partial_y]$.

We see that this expression diverges as $r \to 0$. Therefore, we have to renormalize the parameters μ, λ, and κ as follows:

$$\mu \to r^2\tilde{\mu}, \qquad \lambda \to r^2\tilde{\lambda}, \qquad \kappa \to \tilde{\kappa}.$$

From this, we see that the κ parameter is already properly normalized. After doing this, we obtain the following expression for the Lagrangian functional:

$$L[\varphi, \dot{\varphi}](t) = \int \mathrm{d}\boldsymbol{x}\, \frac{1}{2}\left[\tilde{\mu}\dot{\varphi}^2 - \tilde{\lambda}\varphi^2 - \tilde{\kappa}(\boldsymbol{\partial}\varphi)^2\right].$$

$\boxed{2.}$ To canonically normalize the field, we choose:

$$\varphi \to \frac{1}{\sqrt{\tilde{\kappa}}}\phi$$

and obtain the Lagrangian in the following form:

$$L[\phi, \dot{\phi}](t) = \int \mathrm{d}\boldsymbol{x}\left[\frac{\tilde{\mu}}{2\tilde{\kappa}}\dot{\phi}^2 - \frac{\tilde{\lambda}}{2\tilde{\kappa}}\phi^2 - \frac{1}{2}(\boldsymbol{\partial}\phi)^2\right].$$

Now we define the action functional:

$$S[\phi] := \int \mathrm{d}t\, L[\phi, \dot{\phi}](t) \equiv \int \mathrm{d}x^\mu\, \mathcal{L}(\phi, \partial_\mu\phi),$$

where $\mathcal{L}$ is the Lagrangian density. By varying the action:

$$\delta S[\phi] = \int \mathrm{d}x^\mu \delta\mathcal{L} = \int \mathrm{d}x^\mu \left[\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi)\right] = \int \mathrm{d}x^\mu \delta\phi\left[\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right],$$

where we have used the periodic boundary conditions to eliminate the boundary terms, and by setting $\delta S[\phi] \overset{!}{=} 0$, we obtain the Euler-Lagrange equation

$$\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = 0.$$

Using the previously obtained Lagrangian density

$$\mathcal{L} = \frac{\tilde{\mu}}{2\tilde{\kappa}}\dot{\phi}^2 - \frac{\tilde{\lambda}}{2\tilde{\kappa}}\phi^2 - \frac{1}{2}(\boldsymbol{\partial}\phi)^2,$$

and the following relations

$$\frac{\partial \mathcal{L}}{\partial \phi} = -\frac{\tilde{\lambda}}{\tilde{\kappa}}\phi, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_j \phi)} = -\partial_j \phi, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_t \phi)} = \frac{\tilde{\mu}}{\tilde{\kappa}}\partial_t \phi,$$

we derive the equations of motion for the field ϕ

$$\frac{\tilde{\mu}}{\tilde{\kappa}}\partial_t^2 \phi + \partial^j \partial_j \phi + \frac{\tilde{\lambda}}{\tilde{\kappa}}\phi = 0,$$

where $\partial^j \partial_j \equiv -\Delta \equiv -\sum_j \partial_j^2$.

$\boxed{3.}$ From the equations of motion for the field ϕ, we can clearly identify the speed of sound c and the mass m:

$$\frac{1}{c^2} = \frac{\tilde{\mu}}{\tilde{\kappa}} \qquad \text{and} \qquad m^2 = \frac{\tilde{\lambda}}{\tilde{\kappa}}.$$

$\boxed{4.}$ By setting the speed of sound $c = 1$, we can rewrite the equations of motion into a simpler form:

$$\left(\partial^\mu \partial_\mu + m^2\right)\phi = 0,$$

which resembles the Klein-Gordon equation.

The solution of the Klein-Gordon equation can be expressed in terms of plane waves. A general solution can thus be written in the following form using the Fourier transform (which corresponds to the discrete Fourier series in the lattice case):

$$\phi(\boldsymbol{x}, t) = \frac{1}{(2\pi)^2}\int \mathrm{d}\boldsymbol{p}\left[c(\boldsymbol{p})\exp\left[i(\boldsymbol{p}\cdot\boldsymbol{x} - E(\boldsymbol{p})t\right] + c^*(\boldsymbol{p})\exp\left[-i(\boldsymbol{p}\cdot\boldsymbol{x} - E(\boldsymbol{p})t\right]\right],$$

where $E(\boldsymbol{p})$ is defined using the relativistic dispersion relation:

$$E^2(\boldsymbol{p}) = p^2 + m^2.$$

By identifying the terms in the exponential of the solution for the field with the solution for the discrete lattice, we get:

$$\omega_{jk} \leftrightarrow E(\boldsymbol{p}),$$

and using $x_i \leftrightarrow jr$, we obtain:

$$p_i \leftrightarrow \frac{2\pi m}{Nr} = \frac{2\pi m}{L}.$$

Also, $c_{jk} \leftrightarrow c(\boldsymbol{p})$.

Using these relations, we can substitute them into the dispersion relation for the discrete case:

$$\begin{aligned} \omega_{mn}^2 &= \frac{\lambda}{\mu} + \frac{4\kappa}{\mu}\sin^2\left(\frac{\pi m}{N}\right) + \frac{4\kappa}{\mu}\sin^2\left(\frac{\pi n}{N}\right) \\ &\updownarrow \\ E^2(\boldsymbol{p}) &= \lim_{r\to 0}\left[\frac{\tilde{\lambda}}{\tilde{\kappa}}\frac{\tilde{\kappa}}{\tilde{\mu}} + \frac{4\tilde{\kappa}}{\tilde{\mu}}\frac{1}{r^2}\sin^2\left(\frac{p_x r}{2}\right) + \frac{4\tilde{\kappa}}{\tilde{\mu}}\frac{1}{r^2}\sin^2\left(\frac{p_y r}{2}\right)\right] \\ &= m^2 + p_x^2 + p_y^2 = m^2 + p^2, \end{aligned}$$

where we have used the fact that the speed of sound is equal to one. Therefore, we have shown that both dispersion relations are, in a sense, equivalent.